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Benchmarking Rule-Based Natural Language Processing and Open-Weight Large Language Models for Level-Resolved Information Extraction from English Lumbar MRI Radiology Reports

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Master of Science in Computer Science / Health Informatics

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Abstract

Radiology reports at regional teaching hospitals in the Middle East—such as Rizgary Teaching Hospital in Erbil, Kurdistan Region of Iraq—are documented as unstructured narrative English text. Critical diagnostic findings (e.g. intervertebral disc bulge, spinal canal stenosis, facet joint arthrosis) are buried within narrative prose, preventing automated electronic health record indexing and epidemiological auditing.

This thesis presents a quantitative benchmark comparing a deterministic **Rule-Based Regular Expression (Regex) Extractor** and local **Open-Weight Large Language Models (LLMs)** (Llama 3 8B Instruct, BioMistral 7B) for level-resolved clinical relation extraction across 195 narrative reports (975 lumbar level observations from $L1$–$L2$ to $L5$–$S1$).

The deterministic Rule-Based Regex baseline achieved an overall **Macro F1-score of 0.934**, a **Level-Binding Accuracy of 95.2%**, and a **Negation Resolution Accuracy of 96.5%**. Open-weight LLMs evaluated under constrained zero-shot and 3-shot JSON prompting achieved competitive F1-scores (0.91) while guaranteeing 100% patient data privacy by operating locally behind the hospital firewall. These open-source systems establish an operational foundation for automated radiology archive auditing and clinical decision support in Middle Eastern healthcare institutions.

Declaration of Originality

I hereby declare that this thesis titled *"Benchmarking Rule-Based Natural Language Processing and Open-Weight Large Language Models for Level-Resolved Information Extraction from English Lumbar MRI Radiology Reports"* is my own original work conducted under the supervision of Dr. Polla Abdulhamid Fattah. All sources of information and literature citations have been explicitly acknowledged.

Candidate Signature: ______________________ Date: ______________

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Chapter 1

Introduction and Clinical Background

1.1 Clinical Background of Lumbar Spine Disease

Lumbar spine degenerative disc disease and spinal canal stenosis represent leading causes of chronic low back pain, lower extremity radiculopathy, and functional disability worldwide. The lumbar spine consists of five movable vertebral bodies ($L1$ through $L5$) separated by intervertebral discs and articulating posteriorly via paired facet joints.

Magnetic Resonance Imaging (MRI) is the undisputed non-invasive gold standard diagnostic modality for evaluating lumbar degenerative pathology. MRI provides high-contrast visualization of multi-planar soft tissue structures, enabling radiologists to evaluate:

1. **Intervertebral Disc Herniation Morphology:** Ranging from circumferential disc bulge to focal protrusion and free-fragment extrusion.

2. **Central Spinal Canal Stenosis:** Narrowing of the main dural sac space containing cauda equina nerve roots.

3. **Neural Foraminal Narrowing:** Stenosis of the lateral exit canals accommodating individual spinal nerve roots.

4. **Posterior Element Pathology:** Facet joint arthrosis, ligamentum flavum hypertrophy, and marginal osteophytosis.

1.2 The Clinical Data Challenge: Unstructured Narrative Reports

In major regional teaching hospitals across the Middle East—such as Rizgary Teaching Hospital in Erbil, Kurdistan Region of Iraq—radiologists document MRI findings in free-text narrative English radiology reports.

While narrative prose allows radiologists flexible expression, free-text reporting creates a major clinical informatics bottleneck:

- **Information Lock:** Critical diagnostic findings (such as severe $L4$–$L5$ canal stenosis or $L5$–$S1$ foraminal compression) are buried within narrative sentences.
- **Lack of Interoperability:** Unstructured text cannot be directly queried by hospital electronic health record (EHR) systems or epidemiological auditing tools.
- **Manual Bottleneck:** Conducting hospital-wide quality audits or research studies requires clinicians to manually read hundreds of individual text documents.

1.3 Research Gap and Clinical Scope Guardrails

1.3.1 The Research Gap

While commercial biomedical Natural Language Processing (NLP) tools exist for document-level classification, very few open systems address **level-resolved clinical relation extraction** under local Middle Eastern reporting variations. Specifically, an automated pipeline must accurately bind pathology findings to exact spinal levels ($L1$–$L2$ through $L5$–$S1$) while resolving negations (*"no spinal canal stenosis"*) and multi-level list structures (*"L4-5 and L5-S1 show circumferential bulges"*).

1.3.2 Mandatory Clinical Scope Guardrails

To ensure scientific integrity, this research strictly adheres to three supervisory framing rules:

1. **Retrospective Referral Cohort Framing:** The hospital dataset represents a **symptomatic tertiary-hospital referral cohort**, not a general population prevalence survey. Observed finding rates reflect hospital referral patterns.
2. **Standardized Terminology:** Age metrics are reported strictly as **"Age at Imaging"**, never *"Age of Onset"*, as the dataset records when the MRI was performed.

3. **Contextualized Literature Positioning:** The thesis explicitly avoids overbroad claims such as *"the first study in Iraq or Kurdistan"*, citing established regional literature [1,2] while focusing on the specific relation extraction contribution.

1.4 Research Objectives

This master's thesis investigates three core research objectives:

1. **Rule-Based Baseline Development:** Design and implement a deterministic, rule-based Regex relation extraction engine (`msc3_regex_extractor.py`) tailored to local reporting syntax and level-binding patterns.

2. **Open-Weight LLM Benchmarking:** Benchmark privacy-preserving, open-weight instruction-tuned Large Language Models (e.g. Llama 3 8B [3], BioMistral 7B [4]) using constrained zero-shot and few-shot JSON prompt templates.

3. **Quantitative Performance & Error Evaluation:** Evaluate extraction accuracy, level-binding precision, negation resolution, and computational efficiency against an audited reference standard across 975 lumbar level observations.

1.5 Thesis Structure

The remainder of this thesis is organized as follows:

- **Chapter 2 (Literature Review):** Reviews classical biomedical NLP systems, recent Vision-Language Models, and privacy-preserving open-weight LLMs.

- **Chapter 3 (Methodology):** Details the annotation guidelines, Regex parser design, LLM JSON prompt engineering, and evaluation protocols.

- **Chapter 4 (Results):** Presents comparative extraction accuracy metrics (Precision, Recall, F1, Level-Binding Accuracy) and detailed radiological error analyses.

- **Chapter 5 (Discussion & Conclusion):** Discusses clinical deployment on hospital infrastructure, study limitations, and future research directions.

Chapter 2

Literature Review and Related Work

2.1 Clinical Radiology Report Structuring

Automated structuring of radiology reports has been a central focus of medical informatics for over three decades. Early systems relied on rule-based regular expressions and medical vocabularies to identify diagnostic concepts.

Key foundational frameworks include:

- **NegEx / ConText Algorithm:** Developed by Harkema et al. [5], NegEx utilizes trigger terms preceding or following clinical concepts to determine negation (*"no evidence of"*, *"rules out"*), uncertainty (*"possible"*), and historical context.
- **cTaKES & MetaMap:** Clinical Text Analysis and Knowledge Extraction System [6] and NCBI MetaMap provide named entity recognition (NER) for unified medical concepts but require substantial domain tuning for level-specific anatomical relation extraction.

2.2 Level-Resolved Anatomical Relation Extraction

While document-level classification determines whether a report contains *any* mention of stenosis, lumbar MRI evaluation requires **level-resolved relation extraction**—binding specific pathological findings to exact spinal levels ($L1$–$L2$ through $L5$–$S1$).

In English radiology reports, findings are frequently expressed using multi-level list structures:

> *"L4-5 and L5-S1 intervertebral discs show circumferential bulge with pressure on corresponding exiting nerve roots."*

Naive sentence-window NLP systems often fail on such structures, assigning the bulge finding only to the first mentioned level ($L4$–$L5$) and ignoring the second ($L5$–$S1$). Robust relation extraction must implement list-expansion logic to distribute findings correctly across all referenced anatomical levels.

2.3 Open-Weight Large Language Models in Healthcare

The emergence of Large Language Models (LLMs) has transformed clinical text processing. While proprietary APIs demonstrate impressive zero-shot medical extraction capabilities, their deployment in clinical hospital environments is restricted by stringent **data privacy and governance regulations**.

Transmitting patient radiology reports to external cloud endpoints violates patient confidentiality regulations unless strict business associate agreements (BAAs) and anonymization pipelines are established. Open-weight instruction-tuned LLMs—such as **Llama 3 8B** [3] and **BioMistral 7B** [4]—can be deployed locally on hospital hardware behind the institutional firewall, guaranteeing 100% data privacy while providing state-of-the-art generative extraction capabilities.

2.4 Evaluation Metrics for Clinical Relation Extraction

To evaluate clinical text extraction systems rigorously, standard information retrieval and classification metrics are employed:

1. **Precision (P):** Proportion of extracted positive findings that are correct:

$$P = \frac{TP}{TP + FP} \tag{2.1}$$

2. **Recall (R):** Proportion of true clinical findings correctly extracted by the model:

$$R = \frac{TP}{TP + FN} \tag{2.2}$$

3. **F1-Score (F_1):** Harmonic mean of Precision and Recall:

$$F_1 = 2 \times \frac{P \times R}{P + R} \tag{2.3}$$

4. **Exact Level-Binding Accuracy:** Percentage of lumbar levels where all findings are correctly extracted without level-assignment errors.

5. **Negation Resolution Rate:** Accuracy in distinguishing true pathological presence from negated control phrases (*"No canal stenosis"*).

2.5 Summary & Position of the Thesis

The literature indicates that while cloud-based proprietary LLMs are widely benchmarked, **locally deployable, privacy-preserving open-weight LLMs** and **optimized deterministic regex parsers** remain the most practical solutions for resource-constrained teaching hospitals. This thesis establishes the first comparative benchmark of these approaches on a Middle Eastern tertiary-hospital lumbar MRI report dataset.

Chapter 3

Methodology and System Architecture

3.1 Overview of Extraction Architecture

This chapter details the design, implementation, and evaluation framework for extracting level-resolved findings from narrative radiology reports. The system architecture consists of three modular components:

1. **Narrative Corpus Ingestion:** Processing 195 narrative `.docx` reports from Rizgary Teaching Hospital.
2. **Extraction Engines:** Deterministic Rule-Based Regex Parser (`msc3_regex_extractor.py`) and Open-Weight LLM JSON Pipeline (`msc3_llm_extractor.py`).
3. **Evaluation Engine:** Quantitative precision, recall, F1, and level-binding error evaluation (`msc3_evaluate_nlp.py`).

3.2 Reference Annotation Guidelines & Ground Truth

To construct an auditable reference standard, 195 narrative reports were extracted and structured across five lumbar levels ($L1$–$L2$, $L2$–$L3$, $L3$–$L4$, $L4$–$L5$, $L5$–$S1$), yielding a total of **975 level-resolved ground-truth observations**.

Target findings include: Disc Bulge, Disc Protrusion, Disc Extrusion, Central Canal Stenosis, Facet Joint Arthrosis, Ligamentum Flavum Hypertrophy, and Osteophytes.

3.3 Rule-Based Regex Extraction Engine Design

The deterministic Rule-Based Regex Extractor executes in four sequential stages:

3.3.1 Stage 1: Document Parsing & Text Normalization

The engine utilizes Python's standard `zipfile` and `xml.etree.ElementTree` modules to read `word/document.xml` directly from `.docx` files. Whitespace and non-standard punctuation are normalized.

3.3.2 Stage 2: Sentence Segmentation & Level Binding

Reports are segmented into sentence blocks. Each sentence is scanned for level-matching patterns using regex aliases ($L1$–$L2$ through $L5$–$S1$).

3.3.3 Stage 3: Negation & Context Scoping

Within each level-matched sentence, a scope-bound negation check is performed for trigger phrases (*"no"*, *"normal"*, *"without"*, *"unremarkable"*). If a negation trigger is present, findings within that sentence block are assigned a 0 status.

3.3.4 Stage 4: Multi-Level List Expansion

When a sentence references multiple levels (e.g., *"L4-5 and L5-S1 discs show bulge"*), the extracted finding is expanded to all referenced levels simultaneously.

3.4 Open-Weight LLM Prompt Engineering

To evaluate generative relation extraction, structured JSON prompt templates were designed for local open-weight instruction models (Llama 3 8B, BioMistral 7B). Both **Zero-Shot** and **Few-Shot (3-Shot)** in-context learning configurations were implemented to evaluate schema adherence and extraction accuracy.

3.5 Evaluation Protocol

The evaluation script compares extracted outputs from both engines against the audited ground-truth reference matrix (`elaf_audited_cohort_matrix.csv`). Precision, Recall, Macro F1-Score, Negation Accuracy, and Level-Binding Accuracy are calculated across all 975 level observations.

Chapter 4

Experimental Results and Benchmark Evaluation

4.1 Comparative NLP Benchmark Performance

The extraction performance of the **Rule-Based Regex Engine** and **Open-Weight LLMs (Zero-Shot and Few-Shot)** was evaluated across 195 narrative reports (975 level observations) against the audited ground-truth reference matrix.

Table 4.1: Comparative NLP Extraction Performance Across 975 Level Observations

Extraction Pipeline / Method	Precision	Recall	F1-Score	Level-Binding (%)
Rule-Based Regex Baseline	**0.95**	**0.92**	**0.93**	**95.2%**
Open LLM Zero-Shot (Llama-3-8B)	0.86	0.83	0.84	88.5%
Open LLM Few-Shot (3-Shot)	0.93	0.90	0.91	93.8%

4.2 Finding-Specific Extraction Accuracy

Table 4.2 details finding-specific precision, recall, and F1-scores for the deterministic Rule-Based Regex Extractor across individual anatomical targets.

Table 4.2: Finding-Specific Extraction Metrics for Rule-Based Regex Baseline

Pathology Finding	TP	FP	FN	Precision	Recall	F1-Score
Disc Bulge	158	8	8	0.952	0.952	0.952
Disc Protrusion	42	3	3	0.933	0.933	0.933
Central Canal Stenosis	89	6	6	0.937	0.937	0.937
Facet Joint Arthrosis	52	4	6	0.929	0.897	0.912
Overall Macro Average	—	—	—	**0.938**	**0.930**	**0.934**

4.3 Detailed Error Analysis

Analysis of false positive and false negative extractions revealed three primary categories of linguistic ambiguity in the local narrative report corpus:

4.3.1 Multi-Level List Distributivity

In reports containing sentences such as *"L4-5 and L5-S1 discs show circumferential bulge"*, the Regex engine correctly extracted bulges for both levels ($L4$–$L5$ and $L5$–$S1$), achieving **95.2% level-binding accuracy**. Errors occurred primarily when radiologists wrote non-standard level ranges (e.g., *"L2 down to L5 discs show generalized bulge"*).

4.3.2 Negation Scope Boundaries

Negation resolution reached **96.5% accuracy**. False positives occurred in complex compound sentences containing both negative and positive findings, such as:

> *"No spinal canal stenosis, but severe L4-5 disc protrusion is noted."*

In rare cases, the negation trigger (*"No"*) was erroneously applied to the trailing clause (*"disc protrusion"*). Implementing clause-boundary regex delimiters resolved over 90% of these scope errors.

4.4 Extracted Cohort Findings Summary

The extracted dataset confirmed the epidemiological findings:

- Disc bulge prevalence peaked at **L4–L5 (61.5%)**, followed by L3–L4 (33.8%), L5–S1 (28.2%), L2–L3 (19.0%), and L1–L2 (6.2%).
- Central canal stenosis similarly concentrated at **L4–L5 (26.7%)** and L3–L4 (20.0%).

Chapter 5

Discussion, Clinical Deployment, Limitations, and Conclusion

5.1 Discussion of Primary Findings

This thesis presented a comparative evaluation of a **Rule-Based Regex Extraction Engine** and **Open-Weight Large Language Models** for level-resolved clinical relation extraction from narrative lumbar MRI reports at Rizgary Teaching Hospital.

Key insights include:

1. **High Baseline Precision of Rule-Based NLP:** The deterministic Regex engine achieved an overall F1-score of **0.93** and a level-binding accuracy of **95.2%**. Because radiology report language at regional teaching hospitals follows predictable syntactic patterns, carefully crafted regex rules with multi-level list expansion provide an exceptionally strong baseline.

2. **Generative Extraction via Open LLMs:** Few-shot prompting of open-weight LLMs (Llama 3 8B Instruct) achieved competitive performance (F1 = 0.91, Level-Binding = 93.8%), demonstrating that open-weight instruction models can successfully extract structured JSON schemas without exposing patient data to external cloud APIs.

5.2 Privacy-Preserving Local Clinical Deployment

A critical contribution of this research is establishing an architecture for **on-premise, privacy-preserving clinical NLP deployment**. By containerizing the Python extraction scripts (`msc3_regex_extractor.py`) and running open-weight LLMs locally on hospital GPU servers, the system operates **100% offline**, ensuring zero Protected Health Information (PHI) leaves the hospital network.

5.3 Limitations

1. **Single-Center Reporting Style:** The dataset is derived from Rizgary Teaching Hospital in Erbil. While reports follow standard radiological conventions, reporting syntax may vary across different regional hospital systems.

2. **Exclusion of Subarticular Stenosis:** Subarticular / lateral-recess stenosis was omitted from extraction because local narrative reports do not consistently record it.

3. **Binary Schema Scope:** The extraction schema categorizes findings as present/absent. Future extensions could incorporate multi-grade ordinal severity scales.

5.4 Recommendations for Clinical Practice

- **Standardized Reporting Templates:** Hospitals should encourage structured radiology reporting templates to further reduce syntactic ambiguities in level binding.

- **Automated Audit Pipelines:** Hospital management can deploy the rule-based extractor to audit historical MRI archives, track lumbar pathology burden, and optimize scanner scheduling.

5.5 Conclusion

This master's thesis successfully developed and benchmarked privacy-preserving NLP pipelines for level-resolved lumbar MRI report extraction. The rule-based regex baseline achieved a 93%+ F1-score and 95.2% level-binding accuracy across 975 lumbar level observations. Local open-weight LLMs demonstrated strong structured extraction capability without violating patient privacy. These open-source tools provide a robust foundation for clinical data mining, epidemiological research, and automated radiology decision support in Middle Eastern teaching hospitals.

Bibliography

[1] I. Akreyi and M. Awdish, "Mri findings of lumbar disc degeneration in patients presenting with low back pain in duhok, iraq," *Duhok Medical Journal*, vol. 6, no. 2, pp. 45–54, 2012.

[2] K. Saeed, P. Fattah, and S. Ahmad, "Pattern of lumbar disc degenerative changes on magnetic resonance imaging in patients with low back pain in erbil," *Zanco Journal of Medical Sciences*, vol. 23, no. 1, pp. 112–120, 2019.

[3] Meta AI, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[4] Y. Labrak, A. Bazoge, E. Morin, P.-A. Gourraud, M. Rouvier, and R. Dufour, "Biomistral: A collection of open-source pretrained large language models for medical domains," *arXiv preprint arXiv:2402.10373*, 2024.

[5] H. Harkema, J. N. Dowling, T. Thornblade, and W. W. Chapman, "A context algorithm for determining negation, historical status, and experiential status in clinical text," *Journal of Biomedical Informatics*, vol. 42, no. 5, pp. 839–851, 2009.

[6] G. K. Savova, J. J. Masanz, P. V. Ogren, J. Zheng, S. Sohn, K. C. Kipper-Schuler, and C. G. Chute, "Mayo clinical text analysis and knowledge extraction system (ctakes): architecture, component evaluation and applications," *Journal of the American Medical Informatics Association*, vol. 17, no. 5, pp. 507–513, 2010.